

Smart Canteen Management System

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To

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1. Introduction

1.1.Project Overview

Project Title: Smart Canteen Management System

Team Members:

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Project Duration: 15-12-2025 – 09-01-2026

Instructor: Ms. Nadia Ashfaq

The Smart Canteen Management System is an Android-based application that allows students to view the canteen menu, place food orders, and track order status, while providing the canteen staff with a simple interface to manage orders and update menu items.

1.2. Objectives

The main objective of this project is to automate and streamline the canteen ordering process. The system aims to minimize waiting time, reduce order errors, and improve overall efficiency. It also provides better order tracking and management for canteen staff. The expected outcome is a fully functional Android application that simplifies interaction between students and the canteen.

1.3. Scope

The system will allow users to browse menu items, place orders, and view order status. Canteen staff will be able to manage orders and update menu items. Online payment and delivery features are excluded in this phase to keep the scope manageable. The system will operate within a controlled academic environment.

2. Requirements Analysis

2.1. Functional Requirements

The system shall allow users to register and log in. Users shall be able to view the available menu with prices. The system shall allow users to place food orders and view order status. Admin or canteen staff shall be able to add, update, or remove menu items and manage incoming orders.

2.2. Non-Functional Requirements

The application should provide quick response time for user actions. Basic authentication will be implemented to ensure data security. The interface will be simple and easy to use for both students and staff. The system should be reliable and handle multiple orders efficiently.

2.3. Stakeholders

Primary stakeholders include students, canteen staff, and the university administration. Secondary stakeholders include instructors evaluating the project.

3. System Design

3.1. Architecture

The system will follow the MVC (Model-View-Controller) architecture to separate user interface, business logic, and data handling, ensuring better maintainability and scalability.

3.2. Technologies Used

Frontend: XML (Android UI)

Backend: Java (Android)

Database: SQLite / Room Database

Other Tools: Android Studio, Git

4. Implementation

4.1 Development Process

The project will follow an incremental development methodology. Features will be implemented in small phases, tested individually, and integrated gradually to ensure stability and correctness.

5. Timeline

Phase	Duration	Task
Phase 1	4 days	Requirement Analysis and Planning
Phase 2	7 days	UI Design and Database Design

Phase 3	7 days	Core Feature Implementation
Phase 4	5 days	Testing and Debugging
Phase 5	3 days	Final Documentation and Submission

6. Deployment

6.1. Deployment Environment

The application will be deployed locally on Android devices and emulators for testing and demonstration purposes.

6.2. Limitations

The system does not include online payment integration or real-time delivery tracking. These features may be added in future versions.

7. Conclusion and Future Work

The Smart Canteen Management System aims to provide an efficient and reliable solution for managing canteen operations within a university. Future enhancements may include online payments, user feedback, order history analytics, and cloud-based database integration.